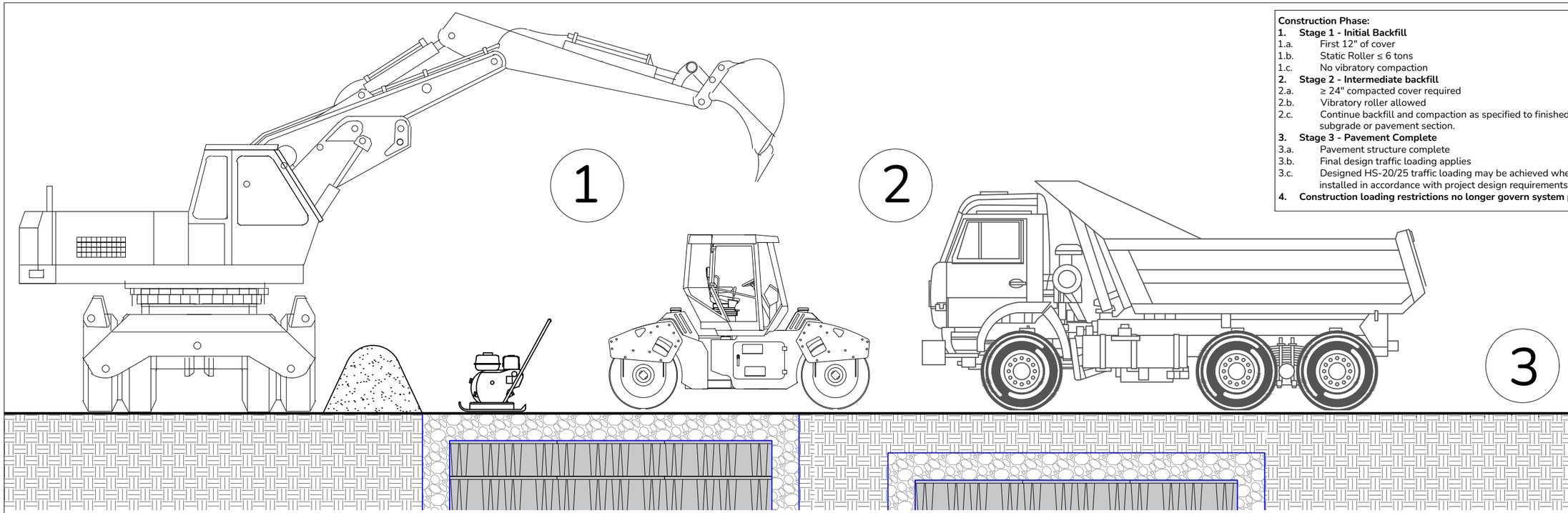


Construction Phase:

1. **Stage 1 - Initial Backfill**
   - 1.a. First 12" of cover
   - 1.b. Static Roller ≤ 6 tons
   - 1.c. No vibratory compaction
2. **Stage 2 - Intermediate backfill**
   - 2.a. ≥ 24" compacted cover required
   - 2.b. Vibratory roller allowed
   - 2.c. Continue backfill and compaction as specified to finished subgrade or pavement section.
3. **Stage 3 - Pavement Complete**
   - 3.a. Pavement structure complete
   - 3.b. Final design traffic loading applies
   - 3.c. Designed HS-20/25 traffic loading may be achieved when installed in accordance with project design requirements.
4. **Construction loading restrictions no longer govern system performance**

Notes:

Fill Placement and Initial Compaction

a. Recommended fill placement method is by tracked excavator operating from outside the AquaCell footprint.
a.a. No compaction equipment shall operate directly on exposed AquaCell units until minimum cover is achieved.
b. Backfill the first 12" layer with an excavator placed outside the footprint and compact with a roller which has an operating weight not exceeding 6 US tons.
c. A minimum of 24" compacted cover measured from the top of the AquaCell units shall be in place before operating a vibratory roller.

Backfill Compaction Standards

d. Backfill should be compacted as required with no loose fill.
d.a. Recommended soil compaction: 90% Standard Proctor for light duty < 2,500 lbs., 95% Standard Proctor for H-10 & HS-20, and 98% Standard Proctor for HS-25 traffic loading.

Construction vs. Final Design Loading

e. The construction loading limits shown apply only during installation, prior to completion of the pavement section.
e.a. These limits should not be interpreted as the final design traffic capacity of the AquaCell system.
e.b. Equipment exceeding the published construction loading limits shall be reviewed by Wavin Technical Support or the engineer of record.
e.c. Final design traffic loads (e.g., HS-20, HS-25) are evaluated based on the completed pavement structure and are significantly greater than construction loading limits.
f. Construction loading conditions and final traffic design loading conditions are not equivalent.
f.a. The final traffic capacity of the AquaCell system is achieved only after the required compacted cover and pavement structure are fully installed.

Proof Rolling and Project Specific Conditions

g. Proof rolling requirements may vary by project.
g.a. Equipment selection and minimum cover requirements should be reviewed based on the project specifications and anticipated construction loads.
g.b. Proof rolling operations conducted with adequate compacted cover (typically ≥ 24" to 36") may be performed using standard geotechnical proof-rolling equipment.
g.c. Consult the engineer of record or Wavin Technical Support if conditions differ.

Managing water better since 1955.
950 Winter Street
Waltham, MA 02451
wavin.us
wavintechsupport@orbia.com

The design details/specifications presented in this document are for informational purposes only and are subject to change without notice. Responsibility for design of any AquaCell system rests soley with the original design engineer of record. For questions, contact: wavintechsupport@orbia.com

Title: Construction Loads Over AquaCell System			
Drawn: HJD	Scale: n.t.s.	Drawing #: AQ-100-20	Rev: **2**
Approved: JS		Date: 9/25/2026	